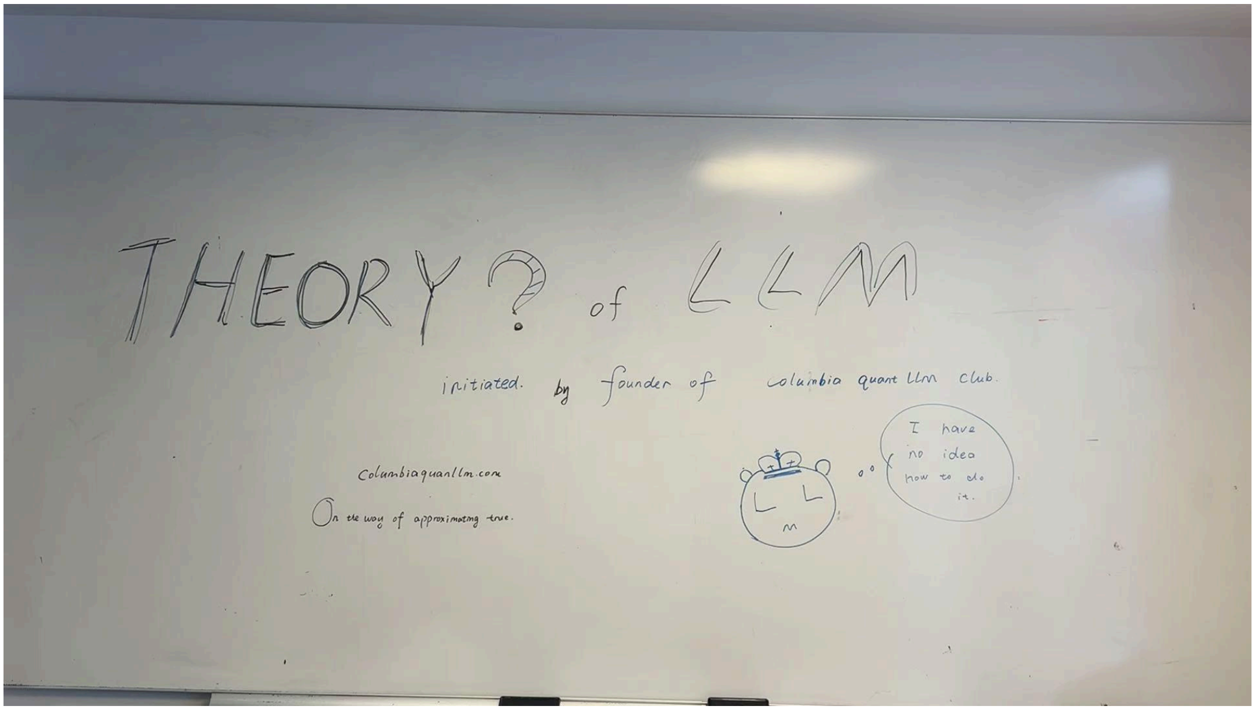
Official webpage: https://autumn-cyber-aka.github.io/columbia-quant-llm.github.io/index.html

CONTINUATION THEORY OF LLM



(Shot in Happening in uris G12 on)

Initiated by founder of columbiaquantllm.com , authorship : crystal ZHANG

Preferred nick name mochi, https://www.linkedin.com/in/crystal-zhangg/

Support by members of CQLLM（abbreviation)

Website built by Jianchen Lyu,https://www.linkedin.com/in/autumnlyu/ 💙
Maintained by ??? feel free to add your name here

Teaching assistant: Hannah Chen

Participants:
From top frontier ai lab, and top hedged fund
Disclaimer, this is not a networking event.
Announcement and Seminar Logistics

Columbia Quant LLM Reading Group

Organized by Columbia Quant LLM columbiaquantllm.com

Mission

This community is centered around large language models (LLMs), with the goal of discussing and sharing frontier research from both theory and empirical perspectives.

Our primary purpose is learning and intellectual exchange. Friendship and collaboration may emerge naturally, but the focus of the group is serious discussion and understanding.

This initiative was originally inspired by the Anthropic Fellowship Program:

https://www.zhihu.com/question/12305646900/answer/1999438034542801397

A guest member of Columbia Quant LLM (MathHeart) shared reflections on participating in the program for the first time.

The structure and long-term form of the community are still under development.

Philosophy

> In front of truth, there is no hierarchy among research directions or research taste. Everyone is equal. Every attempt is simply another step toward approximating truth.

This community is supported by Columbia Quant LLM, with strong inspiration from Stanford open course communities, JQ Investments, and DeepMind.

We welcome:

- Researchers interested in LLMs and AI.
- Students exploring machine learning theory.
- Startup founders and aspiring entrepreneurs.
- Individuals interested in frontier technology.
- Anyone who wishes to understand modern AI more deeply.

Communication

Main communication will be conducted in English.

Our course materials and notes will eventually be made public.

The overall format is inspired by:

- Stanford CS336
- Stanford CS229
- Columbia COMS 6998 (Machine Learning Theory)

Daniel Hsu's course page:

https://www.cs.columbia.edu/~djhsu/coms6998-s25/

Introduction slides:

https://www.cs.columbia.edu/~djhsu/coms6998-s25/intro-slides.pdf

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Community Structure

This is currently a **no-leader discussion community**.

Every participant is encouraged to:

- Share papers they find interesting.
- Present their own work.
- Introduce seminar pages.
- Suggest reading materials.
- Contribute to future schedules.

Every voice is equal and should be heard.

If you have a favorite paper, an interesting question, or a research direction worth discussing, please share it.

You may leave at any time if you feel the group is not useful or interesting to you.

We appreciate your participation and hope that, through collective effort, we can move closer to the truth.

Our official course webpage is currently under development and will be launched soon.

We are also in communication with Columbia faculty members regarding future activities.

Long-Term Goal

Our ultimate goal is to produce research that genuinely improves our understanding of large language models using tools from learning theory, optimization, statistics, complexity theory, and related areas.

We admire ambitious work that combines deep theory with practical relevance.

Our reference standard for research taste is:

ICLR Oral: https://arxiv.org/abs/2210.10749

Our aspiration is not publication for its own sake, but understanding.

The primary purpose of this group remains:

> Learn LLMs.
>
> Understand LLMs.
>
> Approximate truth.

Initial opening Informal Discussion #01

The following reflects my personal and admittedly incomplete perspective. I have not deeply immersed myself in every part of the LLM community.

I strongly encourage everyone to challenge, criticize, improve, or completely replace these ideas.

If someone develops a better version and eventually writes a paper, please remember to credit the original discussion appropriately.

One possible source of hallucination is that a model is often required to produce an answer even when the underlying information is incomplete.

A speculative question is:

> If the information supplied to the model had roughly the same level of detail and structure as the information expected in the output, would the resulting answer become more authentic and less prone to hallucination?

This is only a preliminary thought and should be treated as such.

Background

My own background is primarily in:

- Generative models (dive)
- Deep learning theory(know)
- High-dimensional statistics(know)

I have also worked on:

- Combinatorial optimization(work in)

- Related causal and statistical topics(work in)

Recommended Reading #1

Understanding Deep Learning Requires Rethinking Generalization

https://arxiv.org/abs/1611.03530

ICLR 2017 Best Paper

This is personally one of my favorite examples of academic research taste.

The paper asks an extremely important question:

Why do modern deep networks generalize despite possessing enough parameters to fit random labels?

Many subsequent developments in deep learning theory can be viewed as attempts to answer or refine this question.

Recommended Reading #2

Opening the Black Box of Deep Neural Networks via Information

https://arxiv.org/abs/1703.00810

Over 2000 citations.

This paper remains controversial.

However, controversy is often a sign that a research question is important.

A good research question rarely arrives with a complete answer.

More often, the first answer is incomplete—or even wrong.

Progress comes from repeatedly refining our understanding through collective effort.

That process is precisely what we mean when we say:

> We are approximating the truth.

On Research and Theory

As expressed in one of our Zhihu discussions:

https://www.zhihu.com/question/28256372/answer/128903335713

> In front of truth, there is no hierarchy among research directions or research taste. Everyone is equal. Every attempt is another step toward approximating truth.

Recently I spoke with Tianqi Yang, the STOC Best Student Paper winner.

His view was that many areas of theoretical computer science are deeply connected and can ultimately be understood as different approaches to solving related underlying problems.

My own long-term goal is to become familiar not only with machine learning and machine learning theory, but also with complexity theory and theoretical computer science more broadly.

The prospect is intimidating, but also incredibly exciting.

Perhaps one day we will discover a deeper synthesis between these fields.

On Academic Culture

In what I jokingly call my “PhD Survival Guide for Newcomers” collection, the theory community appears as a group of extraordinarily smart people repeatedly trying ideas, failing, refining, and occasionally producing something that changes the world.

From NTK theory to modern scaling analyses, and especially through work inspired by researchers such as Tengyu Ma and courses like CS229T, we have seen many excellent attempts.

Every serious attempt is a valuable step toward truth.

That, in my view, is one of the highest missions of academia.

Intellectual Credit

Every statement made in this community belongs to its author.

When forwarding, editing, building upon, blogging about, or incorporating another member’s ideas into papers or projects, please provide proper credit.

Please respect:

- Intellectual contributions.
- Academic integrity.
- Columbia’s Code of Conduct.

Future Activities

We plan to organize Zoom meetings and seminar discussions in the future.

The goal is to create an environment where participants can discuss ideas more freely, present favorite papers, and contribute directly to the reading schedule.

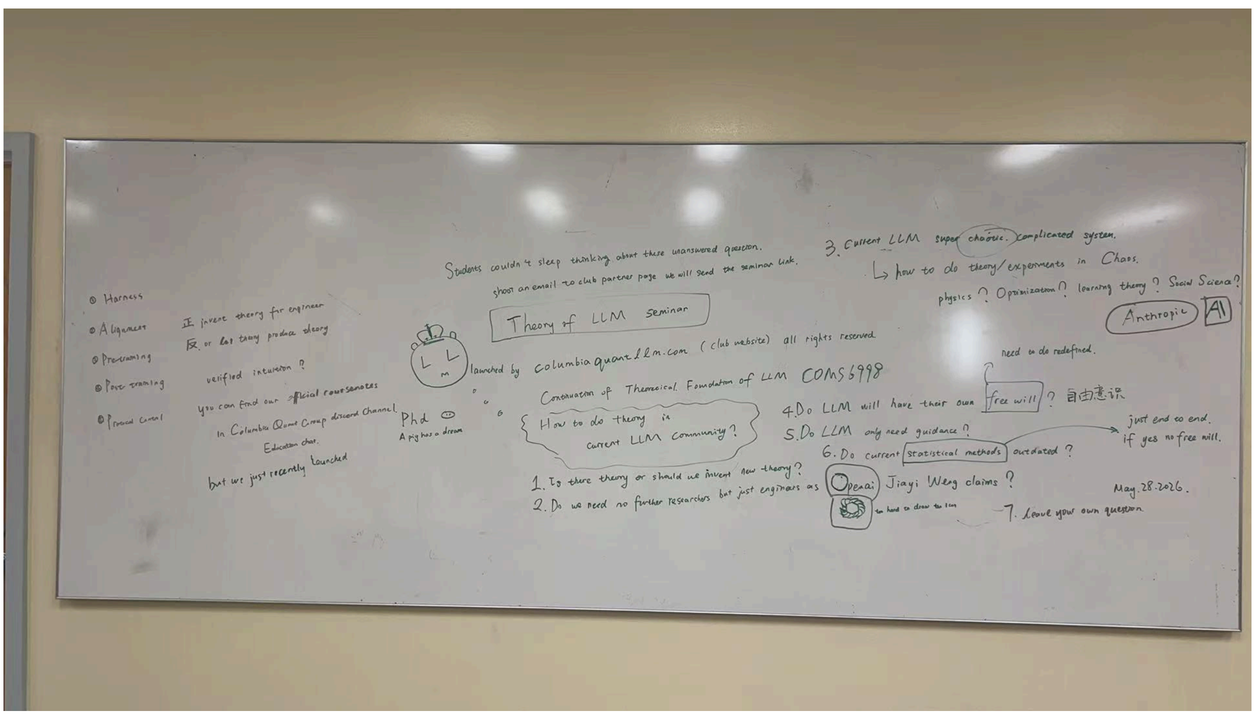
Thank you again for joining.

Welcome to Columbia Quant LLM. columbiaquantllm.com

Fostering next generation AI X QUANT Plus talents and Community.

WHITEBOARD WRITING

May 29, 2026 by Mochi in CS theory or the TA room.



Mar 30, 2026

Third whiteboard post, if interested stay tuned. And shoot an email to t.zhang@columbia.edu

We will talk rigorously for our premium subscription tier group phd student Mingyang DENG world is chaos perspective and Weng Jiayi Openai why we don't need researchers currently, stay tuned.